

Discovery Executive Summary

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EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 3 discovery analyses run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—
2	Code Quality & Complexity Analysis	—
3	Frontend Modernization Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

This is a Laravel 11 + Inertia/React application (PingCRM base) that has been overgrown with a large, deliberately-legacy "IVR" subsystem. Architectural health is **High Risk**: 83 single-action IVR controllers each carry ~759 lines of inline business logic, raw string-concatenated `DB::select` queries, `extract($request->all())`, hard-coded tenant IDs, and `new SomethingGodService()` instantiation — so controllers, persistence, and domain rules are fused into one untestable layer. A Repository tier (12 classes) and a Service tier (12 `*GodService` classes) both exist but are architecturally hollow: the repositories are referenced by zero call-sites (dead abstraction) and the services are god-classes with 45 unrelated public methods, mutable `static` state, and embedded secrets. There is no Dependency Injection wiring at all (`AppServiceProvider` binds nothing; `Model::unguard()` is global), so nothing can be substituted or mocked. On the frontend, 916 components repeat the same pathology in React terms — 374 components issue inline `fetch()` calls to hard-coded URLs, 147 legacy class components mix paradigms, and 8 near-identical 1,101-line "formatter" modules duplicate the same logic. The dominant risks are **change amplification** (one schema or rule change touches dozens of controllers, services, repos and components at once) and **hidden coupling** (reporting reads other domains' operational tables directly), both of which will make any modernization or team scaling extremely costly until bounded contexts, a real service layer, and DI are introduced.

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	~ 683 LOC avg (81 of 91 > 300 LOC; IVR = 759 LOC each)
H2	Missing Service Layer	Controllers accessing repos/models directly	<10	10–20	>20	83 controllers with inline DB/model access + rules
H3	Missing Repository Pattern	Direct DB access points outside repositories	<10	10–20	>20	95 files use DB:: directly; 12 repositories have 0 references
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 cycles observed
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	13 (5 backend helpers + 8 frontend dup formatters)
H6	Direct SQL in Controllers	ORM/repository compliance %	>90%	60–90%	<60%	~ 9% compliant (83 of 91 controllers embed raw SQL)
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0 backend > 1000 LOC, but 12 *GodService @ 45 methods + static state
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	4 (Reports, IvrHub, LoadslvrModuleData, IvrAccountContext)
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	~ 21% (3 of 14 tables shared, no ownership)
H10 (additional)	No Dependency Injection / Service Locator	Services resolved via new	0	1–10	>10	80 new *GodService() ; 0 container bindings
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	avg 139 LOC ; 134 of 522 pages (26%) > 300 LOC
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	374 components with inline fetch() /hard-coded URLs

F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	1 (<code>Hub/Index.tsx</code> = 479 LOC); 134 more in 300–400 band
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	≤2 — components self-fetch into local state
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	147 class components mixed with function components

No additional hotspots beyond H10 were observed.

1.4 Actions Required

Hotspot	Action	Rating	Priority
H1 Fat Controllers	Reduce the 83 IVR controllers to validate-and-delegate; extract logic into Application Services	High Risk	Critical
H2 Missing Service Layer	Introduce per-capability Application Services; move query/workflow assembly out of controllers	High Risk	Critical
H6 Direct SQL in Controllers	Move all <code>DB::</code> queries out of controllers into bound-parameter repositories; add CI grep gate	High Risk	Critical
H10 No Dependency Injection	Bind interfaces in <code>AppServiceProvider</code> , inject services, remove <code>new *GodService()</code> and <code>Model::unguard()</code>	High Risk	Critical
H3 Missing Repository Pattern	Give repositories interfaces + bound queries; route all persistence through them	High Risk	High
H5 Shared Utility Abuse	Replace 5 <code>LegacyIvr*</code> helpers with domain services; collapse 8 duplicate formatter modules into one	High Risk	High
H7 God Classes	Split each <code>*GodService</code> into domain service + repository + application service; remove static state/secret	High Risk	High
F2 Missing Frontend Service Layer	Introduce a typed API client + hooks; replace 374 inline <code>fetch()</code> /hard-coded URLs	High Risk	High
F5 Legacy Component Patterns	Codemod 147 class widgets to hooks; add error boundaries and a ban-class lint rule	High Risk	High
H8 Domain Boundary Violations	Introduce published read models / ACL; forbid cross-context table reads in Reports & Hub	Moderate	Medium

H9 Shared Database Coupling	Assign per-table ownership; serve reporting via an owned read model instead of live joins	Moderate	Medium
F1 Business Logic in Components	Move validation/transform into shared hooks/schema; split 300–400-LOC page templates	Moderate	Medium
F3 God / Oversized Components	Decompose <code>Hub/Index.tsx</code> (479 LOC) and the shared 392-LOC template into widgets + hooks	Moderate	Medium

1.5 Expected Outcomes

- **Testable, reusable business logic** — with Application/Domain Services and constructor DI in place, workflows can be unit-tested with mocked repositories and reused from HTTP, CLI and queued jobs instead of being copy-pasted across 83 controllers.
- **Schema and persistence freedom** — funnelling every `DB::` call through parameter-bound repositories removes the injection holes and lets the team rename/repartition `ivr_*` tables without editing dozens of hand-written query strings.
- **Independent, extractable domains** — bounded contexts with published interfaces and an anti-corruption layer let Queue, Call Routing, Analytics and Reporting evolve (and eventually extract) without silently breaking each other through shared tables.
- **A consistent, maintainable frontend** — a single typed API client plus hooks and function components collapses 374 inline fetches and 147 class widgets into one convention, so endpoint/auth changes are one-file edits and screens stop leaking timers and blanking on errors.
- **Sharply lower change amplification** — de-duplicating the 5 helpers and 8 formatter modules and enforcing CI gates (no `DB::` in controllers, no `React.Component`, duplication check) means a single rule change stops rippling through dozens of near-identical files. {"tftf_ms":7882,"tftf_stream_ms":7697,"time_to_request_ms":6140,"type":"result","duration_ms":491456,"uuid":"07a4177b-031e-4e68-a1fd-2230b3ef721b"}

2. Code Quality & Complexity Analysis

EXECUTIVE SUMMARY

This codebase is a Ping CRM base (Laravel + Inertia/React) onto which a very large "Legacy IVR" surface has been grafted, and that surface is dominated by copy-paste rather than genuine algorithmic complexity. Coverage spans **both layers**: 141 backend PHP files and 1,051 frontend TypeScript/TSX files (1,192 total). The most severe findings are structural, not branch-depth: the frontend contains 8 files over 1,100 LOC and 133 near-identical `LegacyPass2_*.tsx` pages, while the backend has 82 fat IVR controllers at exactly 759 LOC each, 12 `*GodService` classes, and 12 near-identical repositories — pushing estimated overall duplication well above 60%. By contrast, per-method cyclomatic complexity is genuinely low (max ≈ 8) and no single function exceeds ~22 LOC, so those hotspots rate Good. Git history was available (124 commits, 2020–2026); churn and defect-fix frequency on the maintained core are low, but note that the entire high-risk IVR surface arrived in a single bulk commit — so its near-zero churn is deceptive, not reassuring. Additional backend anti-patterns were confirmed: shared mutable `public static` state in all 12 God services and ~4,940 uses of `extract()` for dynamic variable creation.

2.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	~8 (<code>IvrHubController::l</code>
H2	Large Classes	Largest class/file LOC	<300	300–1000	>1000	1,101 LOC (frontend); 759 (backend)
H3	Large Functions	Largest function LOC	<50	50–200	>200	~22 LOC (<code>handleStore</code>
H4	Business Logic Duplication	Duplicated business logic %	<5%	5–10%	>10%	~60% (handlers + workflow per module)
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	~65% (est., manual — no jscpd/phpcpd configured)
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	~0.35/mo (top file, 27 char mo)
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	3 (<code>Contacts/Index</code>)
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	43–71% (maintained core
H9	Global Mutable State (additional)	Shared mutable static/global holders (target 0)	0	1–3	>3	12 (<code>public static \$sharedRuntimeCache</code>)
H10	Dynamic Variable Creation (additional)	<code>extract()</code> /dynamic-var uses (target 0)	0	1–50	>50	~4,940 <code>extract(\$payl</code>

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	18	4.50
Code Churn	25%	15	3.75

Defect Density	20%	30	6.00
Class/Function Size	15%	80	12.00
Business Logic Duplication	10%	92	9.20
Developer Ownership Risk	5%	55	2.75
Hotspot Score	100%		38 / 100

2.5 Actions Required

Hotspot	Action	Rating	Priority
H4 Business Logic Duplication	Consolidate per-module workflows into one <code>IvrModuleService</code> + a shared <code>useIvrRecordForm</code> hook	High Risk	Critical
H5 Duplicate Code (general)	Replace 133 <code>LegacyPass2</code> pages / 8 formatter clones / 12 repos with data-driven components; add <code>jscpd</code> / <code>phpcpd</code> CI gate	High Risk	Critical
H2 Large Classes	Split 82 fat IVR controllers into single-action controllers + services; collapse >1,000-LOC formatter modules	High Risk	High
H9 Global Mutable State	Replace <code>public static \$sharedRuntimeCache</code> with request-scoped/per-tenant cache in all 12 God services	High Risk	High
H10 Dynamic Variable Creation	Replace ~4,940 <code>extract(\$payload)</code> calls with explicit access/DTOs; ban <code>extract()</code> via PHPStan	High Risk	High
H8 Ownership Issues	Add CODEOWNERS for CRM core and assign an owner/deprecation decision for <code>Legacy/**</code> and <code>Pages/Ivr/**</code>	Moderate	Medium

2.6 Expected Outcomes

- **Lower maintenance cost:** consolidating duplicated workflows and files removes an estimated ~60% of redundant code, so each business-rule change is made once instead of across 12–230 sites.
- **Fewer regressions:** a single shared service + form hook eliminates the "fixed here but not there" defect class that dominates the IVR surface today.
- **Faster, safer builds:** deleting 8 oversized formatter modules and ~360 clone components shrinks the frontend bundle and CI time; a duplication gate prevents new clones.
- **Correctness & isolation:** removing shared static state and `extract()` closes cross-tenant leakage and restores static-analysis (Larastan) coverage.
- **Clearer ownership:** CODEOWNERS on the maintained core and an explicit decision on the Legacy/IVR surface reduce coordination overhead and stop dead code from accreting.

Report saved to docs/discovery/02-code-quality-complexity.md (both frontend and backend layers covered). The orchestration UI will convert it to the matching PDF.", "tftt_ms":4694,"tftt_stream_ms":3705,"time_to_request_ms":125,"type":"result","duration_ms":452890,"uuid":"7c0f0161-e69f-4426-b88a-b19daa69a597"}

3. Frontend Modernization Analysis

EXECUTIVE SUMMARY

This is the React/Inertia.js single-page frontend of a PingCRM-derived CRM that has been heavily extended with a synthetic "IVR" telephony surface. The core PingCRM slice (Pages/Contacts, Organizations, Users, Auth, Reports and the 14-component `Shared/` library) is clean, idiomatic React 19 with TypeScript strict mode on. The IVR extension, however, is a large modernization liability: 916 component files, of which ~517 are near-identical duplicates (133 `LegacyPass2_*` page clones, 229 `*Monolith*` components, 147 class-based `.jsx` widgets, 8 byte-identical formatter utilities). The most severe findings are the total absence of an API/service layer — all 874 `fetch()` calls are made inline inside components against an `unauthenticated ivr-legacy/*` route group — plus 375 files that start a 5-second `setInterval` poll with no cleanup (a systemic memory/network leak), 13,701 inline-`style` occurrences with zero design tokens, and a High-severity lodash CVE in a dependency that the frontend does not even import. TypeScript strict is enabled and ESLint is configured, but ESLint is not run in CI, and there are no error boundaries anywhere in the tree.

§3.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	UI Component Duplication	Duplicate components %	<5%	5–10%	>10%	~517 / 916 ≈ 56% near-identical	High R
H2	Legacy Class-Based Components	Modern component adoption %	>90%	70–90%	<70%	84.0% (147 class <code>.jsx</code>)	M
H3	Massive Components	Largest component LOC	<200	200–500	>500	479 LOC (134 files >300)	M
H4	Global State Dependencies	Components reading global state %	<30%	30–60%	>60%	~ 0% (no global store)	G
H5	Complex State Management	Max prop-drilling depth	<3	3–5	>5	≤ 2 (Inertia page props)	G

H6	Weak Frontend Architecture	Feature modules with clean boundaries %	>80%	50–80%	<50%	<20% (API+UI+logic mixed)	High R
H7	Missing Component Inventory	Shared component % of total	>30%	15–30%	<15%	1.5% (14 / 916, no Storybook)	High R
H8	No Design System	Inline-style / magic-value occurrences	0–5	6–20	>20	13,701 inline styles, 0 tokens	High R
H9	Routing Structure Weakness	Protected routes with guards %	100%	80–99%	<80%	100% page routes guarded	
H10	No API Integration Layer	API calls in service layer %	>90%	70–90%	<70%	0% (874 inline <code>fetch</code>)	High R
H11	Poor Data Caching	Data-fetching points with caching %	>70%	40–70%	<40%	0% (no query cache)	High R
H12	Weak Frontend Auth	Token storage + routes guarded	httpOnly + 100%	One gap	Both gaps	httpOnly cookie / data API unguarded	M
H13	Frontend Security Vulnerabilities	XSS-risk + hardcoded secrets count	0 each	1–3 total	>3 total	5+ (2 innerHTML, 2 no-SRI, seeded pw)	High R
H14	Frontend Performance Gaps	Memoization / render optimization	good	some gaps	none + leaks	0 memo, 375 interval leaks	M
H15	Browser Compatibility Gaps	Browserslist + polyfills configured	Both present	One missing	Both missing	polyfills / no browserslist	M
H16	Frontend Code Quality	ESLint in CI + TypeScript strict	Both Yes	One Yes	Both No	strict / ESLint not in CI	M
H17	Technical Debt & Dependencies	Critical/High CVEs found	0	1–3	>3	1 High (lodash) + dead deps	M
H18	(additional) useEffect Cleanup / Leaks	Effects with timers lacking cleanup	0	1–10	>10	375 <code>setInterval</code> w/o cleanup	High R

H19	(additional) Missing Error Boundaries	Error boundaries present	≥1 per area	1 global	0	0 boundaries app-wide	1
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§3.4 Actions Required

Hotspot	Action	Rating	Priority
H10 No API Integration Layer	Move 874 inline <code>fetch</code> calls into a typed <code>services/ivrClient</code> with auth-header/error interceptors	High Risk	Critical
H12 Weak Frontend Auth	Add <code>auth</code> middleware to the <code>ivr-legacy</code> route group; restrict mutations to POST/PUT/DELETE	Moderate	Critical
H13 Security Vulnerabilities	Add SRI to CDN scripts, remove seeded <code>password: 'secret'</code> , sanitize pagination HTML	High Risk	Critical
H18 Interval Leaks	Return <code>clearInterval</code> cleanup in 375 effects (or migrate to React Query polling)	High Risk	Critical
H1 Component Duplication	Consolidate <code>LegacyPass2_*</code> , <code>*Monolith*</code> and <code>legacyFormatters1-8</code> into parameterised units	High Risk	High
H6 Weak Architecture	Extract service layer + presentational components; inject <code>tenantId</code> from auth context	High Risk	High
H7 Missing Inventory	Publish <code>Shared/</code> as <code>@/ui</code> + Storybook; migrate IVR raw controls onto it	High Risk	High
H8 No Design System	Introduce design tokens; codemod 13,701 inline styles to Tailwind classes	High Risk	High
H11 Poor Data Caching	Adopt React Query with <code>staleTime</code> and post-mutation invalidation	High Risk	High
H2 Legacy Class Components	Convert 147 <code>*ClassWidget.jsx</code> to typed function components with a shared hook	Moderate	High
H19 Missing Error Boundaries	Add global + per-feature <code>ErrorBoundary</code> with fallback UI	Moderate	High
H16 Code Quality	Add ESLint CI job; re-enable <code>no-explicit-any</code> ; bring <code>.jsx</code> under strict tsconfig	Moderate	Medium
H17 Technical Debt	Remove unused vulnerable <code>lodash</code> + dead <code>react-router-dom</code> ; upgrade prettier/eslint	Moderate	Medium
H3 Massive Components	Replace unrolled markup with data maps; extract logic to hooks (<code>max-lines</code> rule)	Moderate	Medium
H14 Performance Gaps	Fix interval leaks, memoize heavy tables, add a CI bundle budget	Moderate	Medium
H15 Browser Compatibility	Add <code>.browserslistrc</code> ; bundle polyfills locally; wire Autoprefixer	Moderate	Low

§3.5 Expected Outcomes

- A centralized **services/ivrClient** layer replaces 874 scattered **fetch** calls, giving one place for auth headers, CSRF, base URL, typed responses and consistent error UX — and making the API mockable in tests.
- Guarding the **ivr-legacy** route group and removing GET mutations closes an open, unauthenticated data-and-delete surface that currently sits behind an authenticated page shell.
- Fixing the 375 interval leaks (or moving to React Query polling) eliminates a compounding memory/network leak, stabilising long sessions and cutting idle backend traffic.
- Adopting React Query brings caching, loading/error states and post-mutation invalidation, ending stale-after-save UX and redundant refetching.
- Consolidating ~517 duplicated clones into parameterised components/utilities shrinks the tree, kills behaviour drift, and makes a single fix apply everywhere.
- A documented **@/ui** library + design tokens replaces 13,701 inline styles with a single source of truth for colour and spacing, restoring visual consistency and making rebrands one-line changes.
- Converting 147 class widgets to typed hooks and enforcing ESLint in CI with **strict** typing catches hook-rule and type errors (including the leaks) before they reach **master**.
- Removing the unused, CVE-carrying **lodash** and dead **react-router-dom** clears the one High-severity advisory and reduces attack surface and audit noise to zero.
- Adding error boundaries turns a full-page blank-out on any render throw into a contained, recoverable per-feature fallback.
{"tftf_ms":2446,"tftf_stream_ms":1441,"time_to_request_ms":179,"type":"result","duration_ms":487645,"uuid":"15edec16-e292-4579-b37e-0beb5093972f"}